Mohd Safwan Uddin

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Portfolio — GitHub — LinkedIn

Education

Bachelor of Engineering in Computer Science — MJCET, Hyderabad — Expected 2026
Intermediate (MPC) — Narayana Junior College, Hyderabad — 2020 – 2022

Technical Skills

AI / Machine Learning:	Deep Learning (CNN ,NLP ,LLMs), Anomaly Detection, Feature Engineering, Model Evaluation, Reinforcement Learning.
ML Libraries:	TensorFlow, Keras, scikit-learn, NumPy, Pandas, Matplotlib, HuggingFace.
Languages:	Python (Primary), TypeScript, JavaScript, C, SQL, Solidity.
Backend & Web:	FastAPI, RESTful APIs, JWT Authentication, Microservices.
Databases & Tools:	PostgreSQL, MongoDB, Git, Docker, Postman, Jupyter, Figma.
IoT / Hardware:	ESP32, Arduino, Serial Communication, Sensor Integration, Low-latency Pipelines.

Publications

APEX: Policy-Governed Payment-Gated APIs for Agent Systems — arXiv:2604.02023 — Apr 2026

- Engineered a policy-governed framework for payment-gated APIs in AI Agent systems, achieving **100% mitigation** of replay attacks and invalid token access.
- Implemented a FastAPI-based evaluation pipeline that demonstrated a **27% reduction** in policy-violating expenditures through dynamic token enforcement and idempotent transaction management.

AegisUI: Behavioral Anomaly Detection for Structured UI Protocols — arXiv:2603.05031 — Mar 2026

- Designed an end-to-end anomaly detection framework for AI Agent UI payloads; generated a custom dataset of 4,000 labeled samples across 5 attack families and 5 application domains.
- Benchmarked Isolation Forest, Autoencoder, and Random Forest models; achieved **93.1% accuracy** and **0.952 ROC-AUC**, effectively securing agent-to-user communication channels.

Experience

Machine Learning Intern — Mindalike, Hyderabad — Dec 2025 – Feb 2026

- Developed a feature-engineered user matching system using **TF-IDF vectorization** and **cosine similarity** for high-dimensional interest profiles; integrated directly with live production REST APIs.
- Owned the full ML lifecycle including data preprocessing, feature extraction, and model evaluation in a 3-member agile startup team, contributing significantly to product architectural decisions.

Incubatee — SU Knowledge Hub Foundation, Hyderabad — Aug 2025 – Jan 2026

- Built an end-to-end brain MRI tumor classification pipeline using **CNN, SVM, and Random Forest**; implemented advanced image preprocessing and augmentation to improve model robustness.
- Leveraged **Grad-CAM** explainability heatmaps and 3D volume visualizations to provide clinical interpretability, allowing medical professionals to verify model predictions with visual evidence.

Founder — Zenix (Lovable Shipped Season 1) — Jun 2025 – Aug 2025

- Architected a 3-agent AI orchestration pipeline (layout, styling, code generation) for automated UI creation; achieved a **Top 28 ranking in Asia** (Top 1% globally) among 3,000+ participants in a weekly challenge.
- Deployed the platform using Python and FastAPI, focusing on low-latency agent communication and state-of-the-art prompt engineering for complex UI generation tasks.

Projects

NeuroNav – *Brain MRI Classification Platform* — Live — GitHub

- Developed a full-stack platform for MRI classification; integrated a multi-model inference engine (CNN/SVM) with interactive 3D visualizations and Grad-CAM overlays for diagnostic support.

MIRA – *AI Learning Companion with IoT Integration* — GitHub

- Built a multimodal AI companion using **Gemini AI, Whisper STT, and Neural TTS**; optimized an ESP32 audio loop to achieve sub-2-second end-to-end response latency for screen-free learning.

AuthNexus – *Web3 Decentralized Authentication* — Live — GitHub

- Implemented a hybrid authentication system combining **Ethereum/Polygon** wallet verification (Solidity) with off-chain profiles (Supabase), enhancing user privacy through on-chain signature verification.

Achievements

- Published **2 first-author research papers** on arXiv (cs.AI/cs.CR); featured on Gist.Science with global translations.
- **Winner at TechHack3 Hackathon** — **Finalist in a UI Hackathon** — Incubatee at SU Knowledge Hub.
- Ranked **Top 28 in Asia** out of 3,000+ participants in Lovable Shipped Season 1 .